

Multithreading & JVM Internals - Quick Revision Notes

QUICK REVISION & EXAM CHEAT SHEET

Subject: Java Programming | Chapter: Chapter 3: Advanced Multithreading, Concurrency & JVM Internals | Academic Year: 2026-2027

Thread States & Lifecycle

- * NEW -> RUNNABLE -> BLOCKED/WAITING/TIMED_WAITING -> TERMINATED.
- * wait(), notify(), notifyAll() must be invoked within a synchronized block on the monitor object.
- * Thread.sleep(ms) pauses execution without relinquishing lock ownership.

Concurrency Keywords Cheat Sheet

- * volatile: Guarantees visibility across CPU cores; does NOT guarantee atomicity for compound operations (e.g. i++).
- * synchronized: Guarantees both mutual exclusion (atomicity) and visibility.
- * ReentrantLock: Explicit lock supporting tryLock(timeout), fair scheduling, and multiple Condition variables.

JVM Memory & Garbage Collection

- * Stack stores primitive locals and reference variables; Heap stores all dynamic object allocations.
- * Young Generation: Eden + Survivor S0/S1. Objects surviving generational thresholds are promoted to Old Gen.
- * G1 GC divides the heap into equal-sized regions and prioritizes regions with the most reclaimable garbage.